

# Marriott New Central Runbook

CX Switches

Version 2025.1105

**DRAFT**

**HP E**

## Document Revision History

| Date       | Version   | Author         | Change                                                                                                                                                          |
|------------|-----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/31/2025 | 2025.1031 | Richard Wright | Initial Draft                                                                                                                                                   |
| 11/05/2025 | 2025.1105 | Richard Wright | <ul style="list-style-type: none"><li>• Revised errors</li><li>• Added API elements for components without GUI equivalent</li><li>• Submit for review</li></ul> |

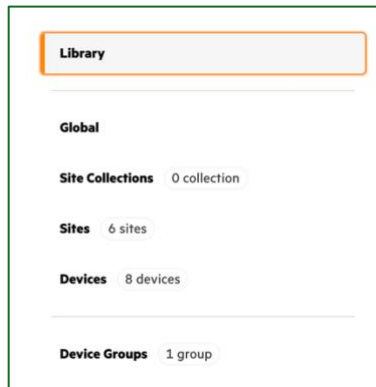
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# CX Switch Configuration Task List

## Navigation

### Left Panel



### Middle Panel



Selected in the example above are **Profiles > Library >**

- From there navigate into the various Profiles
- Select Roles & Policies for Roles and Access Rules
- Select Named Objects for Aliases

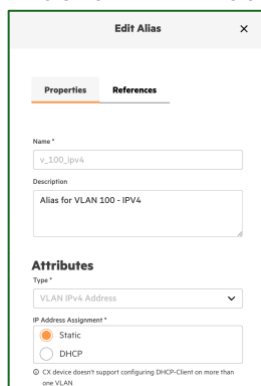
## Create Aliases

### Named Objects > Library > Aliases



In some cases, it may be desirable to create an Alias to be used in a configuration element. This will allow creating a single configuration at a higher level and only need to update the alias value at the device. This will keep the profile linked to the original so that any changes made outside of the alias will be inherited in a lower level.

### Alias for VLAN 100 SVI IP Address:



The Alias will be available where the scope extends

| Name                                                                                                                             | Type              | Assigned Device Function | Assigned Scope |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------|----------------|
|  <b>v_100_ipv4</b><br>Alias for VLAN 100 - IPV4 | VLAN IPv4 Address | -                        | -              |
| <div>Assign</div> <div>Unassign</div>                                                                                            |                   |                          |                |

Set Device Function(s) and assign to Scope(s) as appropriate

Assign Profile

Device Function

☒ Access Switch

☒ Aggregation Switch

☒ Core Switch

☐ Bridge

☐ Campus Access Point

☐ Mobility Gateway

Scopes

Name

Global

Scope Level

Global

Update the Alias at the Device Level and create Override

**Named Objects > Devices > [name of device] > Aliases**

Check the Save as a local profile box

Edit the IP information and Update

Edit Alias

Properties

References

This profile is inherited from GLOBAL. Any modification will break this inheritance and will set the profile as an override.

☒ Save as a local profile

Name \*

v\_100\_ipv4

Description

Alias for VLAN 100 - IPV4

Attributes

Type \*

VLAN IPv4 Address

IP Address Assignment \*

☒ Static

☐ DHCP

CX device doesn't support configuring DHCP-Client on more than one VLAN

IPv4 Address/Prefix \*

10.4.3.130/25

Profiles

Roles & Policies

Named Objects

P5-CX6100

>

Aliases

Access Switch

Q

Search

Create Alias

✓

2 Items

| Name                                                                                     | Type              | Inherits From | Assigned Device Function | Assigned Scope | Overrides               |
|------------------------------------------------------------------------------------------|-------------------|---------------|--------------------------|----------------|-------------------------|
| <div><div></div><div>sys_host_name</div></div>                                           | Hostname          | Self          | Access Switch            | TW11KNP03V     | <div><div></div>-</div> |
| <div><div><div></div></div><div>v_100_ipv4</div><div>Alias for VLAN 100 -...</div></div> | VLAN IPv4 Address | Self          | Access Switch            | TW11KNP03V     | <div><div></div>-</div> |

The override will apply at the level where configured. This is a common method to create configuration elements without creating an override for each element at lower levels. Aliases allow the configuration to be consistent throughout and continually inherited.

### *Types of Aliases*

#### Host

IPv4 Address      XXX.XXX.XXX.XXX  
 IPv6 Address      XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX

#### Identity Profile Control

Enable/Disable      ☐

#### Network

IPv4 Subnet      XXX.XXX.XXX.XXX/XX  
 IPv6 Prefix      XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX/XXX

#### Essid

Essid Name\*      <enter essid name>

#### VLAN

VLAN ID Ranges\*      [select from dropdown]

#### Echo Source IP Address

XXX.XXX.XXX.XXX

#### Network Destination

IPv4 Address      XXX.XXX.XXX.XXX  
 IPv6 Address      XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX  
 IPv6 Prefix      XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX/XXX  
 Domain Name      fqdn

**DNS Server List**

Format IPv4 | IPv6  
xxx.xxx.xxx.xxx  
xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx  
VRF Profile (Switch) select from dropdown

**NTP Server List**

Server Address: IPv4 | IPv6 | FQDN  
Address: xxx.xxx.xxx.xxx  
xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx  
iBurst Mode: Checked | Unchecked  
Min Poll Interval: enter value  
Max Poll Interval: enter value  
Prefer: Checked | Unchecked

**VLAN IPv4 Address** Static | DHCP

**VLAN IPv6 Address** Static | DHCP

**IPv4 Address** xxx.xxx.xxx.xxx

**IPv6 Address** xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx

**Nexthop IPv4 Address** xxx.xxx.xxx.xxx

**Nexthop IPv6 Address** xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx

**IPv4 Prefix** enter value

**IPv6 Prefix** enter value

**VLAN Description** enter description

**Switchport**

Access select Access VLAN from dropdown  
Trunk select Native VLAN from dropdown  
Tag Native VLAN enabled | disabled  
Allowed VLANs

**VLAN Voice Enable** check | uncheck

# General CX Switch Settings

## Hostname

Select Profiles > **Device** > System > System Information

Select Hostname and update information under Edit Profile

The screenshot shows the 'Edit Profile' dialog box on the right side of the screen. The main window displays the 'System Information' tab for a device named 'SG03KN508W'. The 'Edit Profile' dialog has fields for 'Hostname', 'System Description', and 'SNMP V3 Engine ID'. The 'Hostname' field is currently empty.

Adds the following to the switch configuration:

```
hostname MARSHASW01-01-MDF
```

## Admin User

Profiles > Library > System > User Administration > Create Profile

The image shows two side-by-side dialog boxes. The 'Create Profile' dialog on the left has fields for 'Name \*' (MyAdminUser), 'Username \*' (MyAdminUser), 'Password \*' (masked), 'Retype Password \*' (masked), and a 'Role \*' dropdown set to 'Admin'. The 'Assign Profile' dialog on the right shows 'Device Function' options: 'Access Switch' (checked), 'Aggregation Switch' (checked), 'Bridge' (unchecked), 'Core Switch' (checked), 'Campus Access Point' (unchecked), and 'Mobility Gateway' (checked). Below this is a 'Scopes' section with a table showing 'Global' for both 'Name' and 'Scope Level'.

Adds the following to the switch configuration:

```
user MyAdminUser group administrators password ciphertext <xxx>
```

[\(return to TOC\)](#)



## Set Timezone

Profiles > Library > System > Switch System Create Profile

**Edit Profile**

**Properties** **References**

**Switch System**  
Manage switch system configuration.

**General**

Name \*  
switch-system-profile-edge

Description  
Switch profile for 4100, 5420, 6000, 6100, 6200, 6300, 6400, 8100, 8360, 8400, 9300 series switches

Location

Contact

Timezone \*  
US/Eastern UTC(-04:00)

Virtual MAC Address  
XXXXXXXXXXXX

Ⓢ Must be configured at the device level.

☐ Disable USB-Port

☒ Enable Unsupported Transceiver

☒ Enable Port Security

☐ AOS-S Specific Parameters

**Assign Profile**

**Device Function**

☒ Access Switch

☒ Aggregation Switch

☒ Core Switch

**Scopes** +

| Name   | Scope Level |
|--------|-------------|
| Global | Global      |

Adds the following to the switch configuration:

```
clock timezone us/eastern
```

[\(return to TOC\)](#)

## Create VRFs

Profiles > Library > Routing & Overlays > VRF > Create Profile

**Create Profile** Cancel X

Name \*

SW\_WIFI\_MGMT

Description

Advanced

☒ Advanced

**Policies**

Policy

Route Distinguisher

AS:NN or IPAddress:NN (A.B.C.D:NN) format

☐ Use VRF for L3 VPN only

**Route Target**

☐ Address Family IPv4 Unicast

☐ Address Family IPv6 Unicast

☐ EVPN

**Assign Profile** X

**Device Function**

☒ Access Switch

☒ Aggregation Switch

☒ Core Switch

**Scopes**

Adds the following to the switch configuration:

```
vrf SW_WIFI_MGMT
```

[\(return to TOC\)](#)

## Management Interface Settings

Profiles > (Library or Device) > Interfaces > Management Interface > Create Profile  
(Applies to devices with Management VRF/Interfaces)

The default uses DHCP and might not need to be changed

If static configuration is needed, make the following changes:

The image displays two side-by-side screenshots of a network configuration interface for a Management Interface. The left screenshot shows the 'Properties' tab with fields for Name (mgmt), Description, Enable Port (checked), P Mode (Static), and IPv4 Address. The right screenshot shows the 'References' tab with fields for IPv6 Default Gateway, Device Specific Parameters (Switch checked), and Switch Parameters (IPv4 DNS Primary and Secondary Servers). Red boxes highlight the 'Static' P Mode section and the 'Switch Parameters' section.

Change to Static/DHCP based on requirement

Create/Use Aliases if desired

The default adds the following to the switch configuration:

```
interface mgmt
  no shutdown
  ip dhcp
```

[\(return to TOC\)](#)

## Enable SSH to VRF(s)

Profiles > (Library or Device) > System > System Administration > Create Profile

**Edit Profile**

my\_sys\_admin

Description

System Administration Profile with SSH enabled for multiple VRFs

**General**

☒ Console Access

Console Login Attempts

Console Lockout Time

Seconds

☒ SSH Access

Encryption

Both

Encryption configuration is not applicable for Switches

Authentication

☒ HMAC\_SHA1

☒ HMAC\_SHA1\_96

☒ HMAC\_SHA2\_256

☒ Disable Weak DH Key Exchange...

Authentication configuration is only applicable to Gateways

VRF

Selected VRF Profiles

☒ All VRF Profiles

☒ wlan\_vrf

☒ default

☒ mgmt

New VRF

Enable Console Access

Enable SSH Access

Select the VRFs from the dropdown

Assign Profile to Device Function(s) & Scope(s)

**Assign Profile**

**Device Function**

☒ Access Switch

☒ Aggregation Switch

☐ Bridge

☒ Core Switch

☐ Campus Access Point

☐ Mobility Gateway

**Scopes**

+

| Name   | Scope Level |
|--------|-------------|
| Global | Global      |

Adds the following to the switch configuration (if multiple VRFs are supported):

```
ssh server vrf default
ssh server vrf mgmt
ssh server vrf wlan_vrf
```

[\(return to TOC\)](#)

## Enable HTTPS Server VRF

Profiles > (Library or Device) > System > System Administration

The screenshot shows the 'System Administration' configuration page. Red boxes highlight the following sections and settings:

- General:**
  - ☒ Console Access
  - Console Login Attempts: [empty field]
  - Console Lockout Time: [empty field] Seconds
  - ☒ SSH Access
  - Encryption: Both (dropdown)
  - Authentication:
    - ☒ HMAC\_SHA1
    - ☒ HMAC\_SHA1\_96
    - ☒ HMAC\_SHA2\_256
    - ☒ Disable Weak DH Key Exchange Algorithms
  - VRF: default (dropdown)
  - ☒ Web Access
  - VRF: default (dropdown)
- Login Session Timeout:** 5 Minutes
- Authentication:**
  - Authentication Type: RADIUS (selected)
  - ☒ Fallback to Local Authentication
  - ☐ Allow Failthrough
- Authentication Server Group:** RADIUS-1-Group
- Device-Specific Parameters:** Gateway (checkbox)

\*Specify VRF in options above to enable different VRFs for access

Adds the following to the switch configuration:

```
cli-session
  timeout 5
aaa authentication login console group RADIUS-1-Group local
aaa authentication login default group RADIUS-1-Group local
aaa authentication login https-server group RADIUS-1-Group local
aaa authentication login ssh group RADIUS-1-Group local
ssh server
https server
```

[\(return to TOC\)](#)

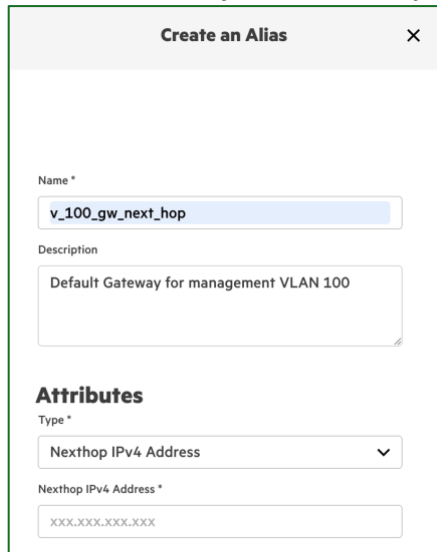
## IP Route/Default Route

Profiles > Library > Routing & Overlays > Static Routing > Create Profile  
(Use with Alias if different 'next-hop address' is required)

Profiles > **Device** > Routing & Overlays > Static Routing  
(Use if configuration is on a single device)

If desired, [create Alias](#)

Profiles > Library > Named Objects > Aliases



**Create an Alias** [X]

Name \*  
v\_100\_gw\_next\_hop

Description  
Default Gateway for management VLAN 100

**Attributes**

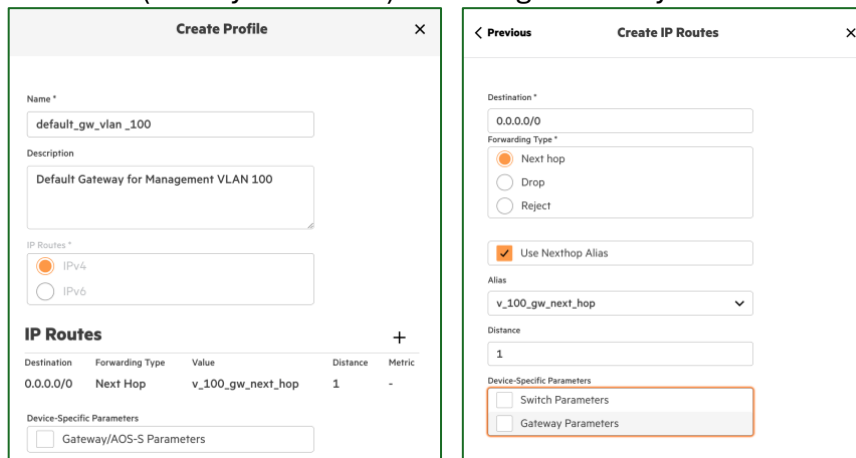
Type \*  
Nexthop IPv4 Address

Nexthop IPv4 Address \*  
XXX.XXX.XXX.XXX

Assign to Device Functions: Core Switch, Aggregation Switch, Access Switch  
Assign to Scope: Global

### Create Route

Profiles > (Library or **Device**) > Routing & Overlays > Static Routing



**Create Profile** [X]

Name \*  
default\_gw\_vlan\_100

Description  
Default Gateway for Management VLAN 100

IP Routes \*  
☒ IPv4  
☐ IPv6

| Destination | Forwarding Type | Value             | Distance | Metric |
|-------------|-----------------|-------------------|----------|--------|
| 0.0.0.0/0   | Next Hop        | v_100_gw_next_hop | 1        | -      |

Device-Specific Parameters  
☐ Gateway/AOS-S Parameters

**Create IP Routes** [X] [Previous]

Destination \*  
0.0.0.0/0

Forwarding Type \*  
☒ Next hop  
☐ Drop  
☐ Reject

☒ Use Nexthop Alias

Alias  
v\_100\_gw\_next\_hop

Distance  
1

Device-Specific Parameters  
☐ Switch Parameters  
☐ Gateway Parameters

Enter next-hop (or use Alias if desired)

## NTP Server

Profiles > Library > System > NTP Server > Create Profile

The image shows three sequential screenshots of a network configuration interface for NTP servers.

- Create Profile:** A form with fields for Name (my\_ntp), Description (NTP Server), and a checkbox for NTP Alias. Below is a section for NTP Server with a table for IP Address and Format. A box icon indicates no data to display. There is a checkbox for Switch Specific Parameters and a section for Switch Parameters with a VRF dropdown menu showing wlan\_vrf, default, and mgmt.
- Add NTP Server:** A form with fields for Server Address (IPv4, IPv6, FQDN), FQDN (pool.ntp.org), iBurst Mode (checked), Minimum Poll Interval (4), Maximum Poll Interval (6), and a checkbox for Prefer.
- Assign Profile:** A form with a warning message about NTP server updates. Below is a section for Device Function with checkboxes for Access Switch, Aggregation Switch, Core Switch, Campus Access Point, and Mobility Gateway. There is also a section for Scopes with a table for Name and Scope Level, showing Global for both.

Adds the following to the switch configuration:

```
ntp server pool.ntp.org minpoll 4 maxpoll 6 iburst
ntp enable
```

\*Note: Devices with multiple VRFs can be configured to select a VRF for NTP

```
ntp server pool.ntp.org minpoll 4 maxpoll 6 iburst
ntp enable
ntp vrf mgmt
```

[\(return to TOC\)](#)

## DNS Servers

Profiles > Library > System > DNS Server > Create Profile

PropertiesReferences

Name \*

Description

Domain Name

☐ DNS Alias

**DNS Servers** +  
GW, AOS-S supports a maximum of 6, 4 DNS servers respectively and CX supports a maximum of 3 DNS servers per VRF.

| IP Address | Format |
|------------|--------|
| 1.1.1.1    | IPv4   |
| 8.8.8.8    | IPv4   |

Assign Profile

**Device Function**

☒ Access Switch  
☒ Aggregation Switch  
☒ Core Switch  
☒ Mobility Gateway

**Scopes** +

| Name   | Scope Level |
|--------|-------------|
| Global | Global      |

Adds the following to the switch configuration:

```
ip dns server-address 1.1.1.1
ip dns server-address 8.8.8.8
```

If desire to set DNS Server to an alternate VRF, select 'Device Specific Parameters' and select the available VRF from the dropdown (or create new)

Edit DNS Server

IP Address Format \*

☒ IPv4☐ IPv6

IP Address \*

Device-Specific Parameters

☒ Switch☐ Gateway

**Switch Parameters**

VRF Profile \*

default

☒ default

View SelectionNew VRF



## Set up Logging

Profiles > Library > System > Syslog Server > Create Profile

The image shows three screenshots of a network configuration interface for setting up a Syslog Server profile.

- Create Profile:** This screen is used to create a new profile. The 'Name' field is set to 'syslog\_server'. The 'Device Type' is set to 'Switch'. The 'Debug Syslog Logging Level' is set to 'Warning'. A table at the bottom shows the profile is associated with 'FQDN/IP Address', 'Logging Level', and 'VRF'. A box icon indicates 'No data to display'.
- Add Logging Server:** This screen is used to add a logging server to the profile. The 'Server Address Format' is set to 'IP Address'. The 'IP Address' is set to '192.168.224.2'. The 'Logging Level' is set to 'Warning'. The 'VRF' is set to 'default'.
- Assign Profile:** This screen is used to assign the profile to a device. The 'Device Function' is set to 'Access Switch', 'Aggregation Switch', and 'Core Switch'. The 'Scopes' table shows the profile is assigned to 'Global' scope.

Adds the following to the switch configuration:

```
logging 192.168.224.2 severity warning
debug destination syslog severity warning
```

[\(return to TOC\)](#)

## SNMP Server Settings

Profiles > Library > System > SNMP > Create Profile

The image shows three screenshots of the SNMP configuration interface:

- Create Profile:** Shows the 'Name' field set to 'XYZ-SNMP', 'Description' as 'SNMP for devices at property', 'Version' set to 'V2C', 'Community Strings' with '678!ZJsaw\*@0091', 'Trap Receivers' with 'CX specific parameter' checked, and 'Switch Parameters' with 'wlan\_vrf', 'default', and 'mgmt' selected.
- Create Trap Receiver:** Shows 'Version' as 'V2C', 'IP Address' as '192.168.224.1', 'Community' as '678!ZJsaw\*@0091', 'Port' as '162', 'Mode' as 'Inform', 'Device-Specific Parameters' with 'Switch' checked, and 'Switch Parameters' with 'wlan\_vrf', 'default', and 'mgmt' selected.
- Assign Profile:** Shows 'Device Function' with 'Access Switch', 'Aggregation Switch', 'Core Switch', 'Campus Access Point', and 'Mobility Gateway' all checked, and 'Scopes' with 'Global' selected.

Adds the following to the switch configuration:

```
snmp-server vrf default
snmp-server community 678!ZJsaw*@0091
snmp-server host 192.168.224.1 inform version v2c community 678!ZJsaw*@0091
```

\*Devices with a single VRF will not have the \*vrf command

## Spanning Tree Settings

Profiles > (Library or Device) > VLANs & Networks > STP > Create Profile

### Spanning Tree Mode: MSTP

#### Default Priority – Edge Switch

The first screenshot shows the 'STP' configuration page. The 'Name' field is 'edge\_stp'. The 'Description' is 'MSTP with default settings for Edge Switches'. The 'Enable STP' checkbox is checked. The 'Device' is set to 'Switch'. The 'Mode' is set to 'MSTP'. The 'Advanced' section shows the 'Default Instance Priority' slider set to 8 (32768). The 'MSTP Instance' table is empty. The second screenshot shows the 'Assign Profile' dialog. The 'Device Function' section has 'Access Switch' and 'Aggregation Switch' checked. The 'Scopes' section has 'Global' selected for both 'Name' and 'Scope Level'.

Adds the following to the switch configuration:

```
spanning-tree
```

#### Higher Priority – Core Switch

The first screenshot shows the 'STP' configuration page. The 'Name' field is 'core\_stp'. The 'Description' is 'MSTP with higher priority for Core Switches'. The 'Enable STP' checkbox is checked. The 'Device' is set to 'Switch'. The 'Mode' is set to 'MSTP'. The 'Advanced' section shows the 'Default Instance Priority' slider set to 4 (16384). The 'MSTP Instance' table is empty. The second screenshot shows the 'Assign Profile' dialog. The 'Device Function' section has 'Core Switch' checked. The 'Scopes' section has 'Global' selected for both 'Name' and 'Scope Level'.

Adds the following to the switch configuration:

```
spanning-tree
spanning-tree priority 4
```

Advanced settings can be added by selecting Advanced

Below example shows enabling traps and selecting region

**Advanced** ^

BPDU Guard Enable Timer(in seconds)

Path Cost Type

☒ LONG

☐ SHORT

Transmit Hold Count

1 (Highest) 10 (Lowest)

6

Traps

☒ Errant BPDU

☒ Loop Guard Inconsistency

☒ New Root

☒ Root Guard Inconsistency

Region

REGION-1

Config Revision

0

The above options add the following to the switch configuration:

```
spanning-tree
spanning-tree trap new-root
spanning-tree trap loop-guard-inconsistency
spanning-tree trap root-guard-inconsistency
spanning-tree trap errant-bpdu
spanning-tree config-name REGION-1
```

## Spanning Tree Mode: RPVST

### Default Priority – Edge Switch

The image shows two screenshots from a network configuration interface. The left screenshot is the 'STP' configuration page for a profile named 'rpvst\_edge\_stp'. It has a description 'RSTP with default settings for Edge Switches'. The 'Enable STP' checkbox is checked. The 'Device' is set to 'Switch'. Under 'Mode', 'RPVST' is selected. Under 'Advanced', 'RPVST Mode' is set to 'Auto RPVST (Include VLANs)' and 'Auto RPVST Priority' is set to 8 (32768). The right screenshot is the 'Assign Profile' dialog. Under 'Device Function', 'Access Switch' and 'Aggregation Switch' are checked. Under 'Scopes', the 'Name' is 'Global' and the 'Scope Level' is 'Global'.

Adds the following to the switch configuration:

```
spanning-tree mode rpvst auto-vlan-enable
spanning-tree
```

### Higher Priority – Core Switch

The image shows two screenshots from a network configuration interface. The left screenshot is the 'STP' configuration page for a profile named 'rpvst\_core\_stp'. It has a description 'RSTP with higher priority for Core Switches'. The 'Enable STP' checkbox is checked. The 'Device' is set to 'Switch'. Under 'Mode', 'RPVST' is selected. Under 'Advanced', 'RPVST Mode' is set to 'Auto RPVST (Include VLANs)' and 'Auto RPVST Priority' is set to 4 (16384). The right screenshot is the 'Assign Profile' dialog. Under 'Device Function', 'Core Switch' is checked. Under 'Scopes', the 'Name' is 'Global' and the 'Scope Level' is 'Global'.

Adds the following to the switch configuration:

```
spanning-tree mode rpvst auto-vlan-enable priority 4
spanning-tree
```

Advanced settings can be added by selecting Advanced

Below example shows enabling traps and selecting region

**Advanced**

BPDU Guard Enable Timer(in seconds)

Path Cost Type

☒ LONG

☐ SHORT

Transmit Hold Count

1 (Highest) 10 (Lowest)

6

Traps

☒ Errant BPDU

☒ Loop Guard Inconsistency

☒ New Root

☒ Root Guard Inconsistency

☐ Ignore PVID Inconsistency

☒ Extend System ID

RPVST MSTP Interconnect-VLAN

1

The above options add the following to the switch configuration:

```
spanning-tree mode rpvst auto-vlan-enable
spanning-tree
spanning-tree trap new-root
spanning-tree trap loop-guard-inconsistency
spanning-tree trap root-guard-inconsistency
spanning-tree trap errant-bpdu
```

[\(return to TOC\)](#)

## Set Loop-protect re-enable timer

Profiles > Library > System > Switch System > Edit/Create Profile

The screenshot shows the 'Switch System' configuration page. The 'Properties' tab is active. The 'General' section includes fields for Name (switch-system-profile-edge), Description (Switch profile for 4100, 5420, 6000, 6100, 6200, 6300, 6400, 8100, 8360, 8400, 9300 series switches), Location, Contact, Timezone (US/Eastern UTC(-04:00)), and Virtual MAC Address. The 'Loop Protect' section is highlighted with a red box and contains the 'Re-Enable Timer' set to 180 seconds, a 'Transmit Interval' slider set to 5 seconds, and a checked checkbox for 'Trap on Loop Detection'.

Adds the following to the switch configuration:

```
loop-protect re-enable-timer 180
loop-protect trap loop-detected
```

[\(return to TOC\)](#)

## Stacking

*Section coming in future update*

[\*\(return to TOC\)\*](#)



# Authentication

## AAA Server

Profiles > Library > Security > Authentication Server > Create Profile

Edit Profile

PropertiesReferences

Name \*  
RADIUS-1

Description

Server Type  
RADIUS

☐ Secure RADIUS

Auth Server Mode  
☒ RADIUS  
☐ RADIUS with CoA (Change of Authorization)  
☐ CoA Only

IP Address/FQDN \*  
192.168.224.1

Shared Secret \*  
\*\*\*\*\*

Retype Shared Secret \*  
\*\*\*\*\*

Authentication Port \*  
1812

Accounting Port \*  
1813

☐ Advanced

Assign Profile

Device Function

☒ Access Switch  
☒ Aggregation Switch  
☒ Core Switch  
☐ Campus Access Point  
☐ Mobility Gateway

Scopes

| Name   | Scope Level |
|--------|-------------|
| Global | Global      |

Adds the following to the switch configuration:

```
radius-server host 192.168.224.1 key ciphertext <cipherkey>
```

[\(return to TOC\)](#)

## AAA Server Group

Profiles > Library > Security > Authentication Server Group > Create Profile

Edit Profile

PropertiesReferences

Name \*

RADIUS-1-Group

Description

Server Type

RADIUS

Authentication Servers \*

RADIUS-1

All items

☒ RADIUS-1

☐ RADIUS-222

☐ default\_radius

☐ radius-proxy-default

☐ sys\_central\_nac

View Selection

New Server

Assign Profile

Device Function

☒ Access Switch

☒ Aggregation Switch

☒ Core Switch

☐ Campus Access Point

☐ Mobility Gateway

Scopes

Name

Scope Level

Global

Global

Adds the following to the switch configuration:

```
aaa group server radius RADIUS-1-Group
server 192.168.224.1
```

[\(return to TOC\)](#)

## Enable Authentication Globally

\*Create [RADIUS Server](#) and [RADIUS Group](#) before enabling Authentication

Profiles > (Library or Device) > System > Switch System

**Switch System**  
Manage switch system configuration.

☐ Save as a local profile

**General**

Name \*  
switch-system-profile-edge

Description  
Switch profile for 4100, 5420, 6000, 6100, 6200, 6300, 6400, 8100, 8360, 8400, 9300 series switches

Location

Contact

Timezone \*  
US/Eastern UTC(-04:00)

Virtual MAC Address  
XXXXXXXXXXXX

☐ Disable USB-Port

☒ Enable Unsupported Transceiver

☒ Enable Port Security

**AAA**

☒ Enable 802.1X

☒ Enable MAC Authentication

MAC Address Format  
None

MAC Radius Auth Method  
None

802.1X Authentication Server Group \*  
RADIUS-1-Group

MAC Authentication Server Group \*  
RADIUS-1-Group

### Assign Function and Scope

**Assign Profile** ✕

**Device Function**

☒ Access Switch

☒ Aggregation Switch

☒ Core Switch

**Scopes** +

| Name   | Scope Level |
|--------|-------------|
| Global | Global      |

Adds the following to the switch configuration:

```
aaa authentication port-access dot1x authenticator
radius server-group RADIUS-1-Group
enable

aaa authentication port-access mac-auth
radius server-group RADIUS-1-Group
enable
```

[\(return to TOC\)](#)

## Login Settings (console, ssh, etc.)

Profiles > (Library or Device) > System > System Administration

The screenshot displays the 'System Administration' configuration page. It is divided into two main panels. The left panel contains fields for 'Name' (switch-system-admin) and 'Description' (System Administration Profile). Below these are sections for 'General', 'SSH Access', 'Authentication', and 'Web Access'. The 'General' section has a 'Console Access' checkbox checked. The 'SSH Access' section has an 'SSH Access' checkbox checked. The 'Authentication' section has 'Authentication Type' set to 'RADIUS' and 'Fallback to Local Authentication' checked. The 'Web Access' section has a 'Web Access' checkbox checked. The right panel contains fields for 'Login Session Timeout' (5 Minutes), 'Login Banner', 'Login Retries', 'Retry Delay Seconds', and 'Authentication Server Group' (RADIUS-1-Group). The 'Authentication Server Group' section has a 'Device-Specific Parameters' checkbox for 'Gateway'.

Name \*

switch-system-admin

Description

System Administration Profile

**General**

☒ Console Access

Console Login Attempts

Console Lockout Time

Seconds

☒ SSH Access

Encryption

Both

Encryption configuration is not applicable for Switches

Authentication

☒ HMAC\_SHA1

☒ HMAC\_SHA1\_96

☒ HMAC\_SHA2\_256

☒ Disable Weak DH Key Exchange Algorithms

Authentication configuration is only applicable to Gateways

VRF

default

VRF configuration is only applicable to CX

☒ Web Access

VRF

default

VRF configuration is only applicable to CX

Login Session Timeout \*

5 Minutes

Login Banner

Login Retries

Retry Delay Seconds

**Authentication**

Authentication Type \*

☐ Local

☒ RADIUS

☐ TACACS+

☒ Fallback to Local Authentication

☐ Allow Failthrough

**Authentication Server Group** +

Name

RADIUS-1-Group

Device-Specific Parameters

☐ Gateway

Adds the following to the switch configuration:

```
cli-session
  timeout 5
aaa authentication login console group RADIUS-1-Group local
aaa authentication login default group RADIUS-1-Group local
aaa authentication login https-server group RADIUS-1-Group local
aaa authentication login ssh group RADIUS-1-Group local
ssh server
https server
```

[\(return to TOC\)](#)

## Authentication Profile

Profiles > Library > Security > AAA Authentication > Create Profile

Name \*

wired-port-auth-no-flood

Description

Port Authentication without Flood for devices (PCs, Computers, Micros, etc.)

Authentication Protocol \*

Concurrent

Client Limit

2

802.1X Authentication Server Group

RADIUS-1-Group

MAC Authentication Server Group

RADIUS-1-Group

Authentication Parameters

☒ Switch Specific Parameters

☐ Allow Flood Traffic

Authentication Mode

Select

Client Auto Logoff

Select

☐ Secure Device Profile

Onboarding Precedence

Device Profile, then AAA

LLDP Support

☒ Allow LLDP Authentication

☐ Chassis MAC

☐ Interface MAC

☒ Allow LLDP Bypass

CDP Support

☒ Allow CDP Authentication

☒ Allow CDP Bypass

☐ Allow CDP Proxy Logoff

Security Violation

Action

Select

Authorization Parameters

☐ Switch Specific Parameters

Accounting

☒ Interim RADIUS Accounting

Accounting Server Group \*

RADIUS-1-Group

802.1X Parameters

Max Authentication Failure

Reauthentication

☒ Switch Specific Parameters

Switch Parameters

☐ Cached Reauthentication

Discovery Period

seconds

EAPOL Timeout

5

seconds

Maximum EAPOL Requests

1

☐ AOS-S Specific Parameters

MAC Parameters

MAC Address Delimiter

MAC Address Case

Max Authentication Failure

0

Reauthentication

☐ Switch Specific Parameters

Update Settings as needed

Optional:  
Enable **Allow Flood Traffic**

The various devices connecting to the wired network may require special settings to allow for authentication when the device does not communicate on the network. This condition keeps the device from MAC auth because the switch doesn't learn of the device and therefore does not authenticate. In this situation, it may be overcome by placing the device (usually a kitchen printer) into the intended VLAN but enable VLAN Flood so that ARP requests can reach the device. When it responds, auth will occur.

Assign to Device Function & Scopes:  
(will be copied to device once referenced in Port Profile)

Assign Profile

Device Function

☒ Access Switch

☒ Aggregation Switch

☐ Bridge

☒ Core Switch

☐ Campus Access Point

☐ Mobility Gateway

Scopes

+

Name

Global

Scope Level

Global

# Security

## Access List

**Roles & Policies** > Library > Security Policies > Network Policies

Create Policy

Name \*

DNSBlock

Description

☒ Switch Specific Parameters

Switch Parameters

☐ Distributed Services Firewall

☒ Access List

Add Rule:

Edit Rule

Description

Source \*

Any

Destination \*

Any

Service/Application \*

Protocol and Port

Protocol

UDP

Port

53

Specify a single port or a dash seperated range of ports

Action \*

Deny

☒ Log

☒ Count

☐ Advanced

Edit Rule

Description

Source \*

Any

Destination \*

Any

Service/Application \*

Any

Action \*

Allow

☐ Log

☐ Count

☐ Advanced

| Profiles Roles & Policies Named Objects Services |             |                     |        |                                                |      |                |
|--------------------------------------------------|-------------|---------------------|--------|------------------------------------------------|------|----------------|
| Library > Security Policies > Network Policies   |             |                     |        |                                                |      |                |
| 0 items                                          |             |                     |        |                                                |      |                |
| Create Policy                                    |             |                     |        |                                                |      |                |
| Name                                             | Type        | Rules               |        | Assigned Device Function                       |      | Assigned Scope |
| DNSBlock                                         | Access List | 2                   |        | Core Switch, Aggregation Switch, Access Switch |      | Global         |
| Source                                           | Destination | Service/Application | Action | Description                                    | DSCP | 802.1P         |
| Any                                              | Any         | UDP Port: 53        | Deny   | -                                              | -    | -              |
| Any                                              | Any         | Any                 | Allow  | -                                              | -    | -              |

## Assign to Device Function and Scope

Assign Profile

Device Function

☒ Access Switch
 ☒ Aggregation Switch
 ☐ Bridge
 ☒ Core Switch
 ☐ Campus Access Point
 ☐ Mobility Gateway

Scopes

+

| Name   | Scope Level |
|--------|-------------|
| Global | Global      |

Adds the following to the switch configuration:

```
access-list ip DNSBlock
  1 deny udp any any eq dns log count
  2 permit any any any

access-list ipv6 DNSBlock
  1 deny udp any any eq dns log count
  2 permit any any any

access-list mac DNSBlock
  2 permit any any any
```

Assign the Access List to interfaces/VLANs as needed

### Interface Example: 1/1/3

Profiles > **Device** > Interfaces > Switch Interface Configuration > select Port

Port \*

1/1/3

Description

MARSHA\_GW01\_WAN

☐ Use Port Profile

☒ Enable Port

Speed Duplex

Auto

MTU

1500

**Policies**

Policy

None

☒ Access List

Access List \*

Outbound

Outbound Access List \*

DNSBlock

Adds the following to the interface configuration:

```
interface 1/1/3
  description MARSHA_GW01_WAN
  no shutdown
  apply access-list ip DNSBlock out
  apply access-list ipv6 DNSBlock out
  apply access-list mac DNSBlock out
  no routing
  vlan access 915
  exit
```



## VLAN Example: 915

Profiles > Library > VLANs & Networks > VLAN > select VLAN ID

PropertiesReferences

VLAN ID \*  
915

Name  
915\_ISP1\_PROVIDER

☐ This configuration is only applicable to Switches

Description

☐ Description Alias

☒ Switch Specific Parameters

Switch Parameters

☒ Enable VLAN

VRF Profile

Policy  
None

☒ Access List

Access List \*  
Outbound

Outbound Access List \*  
DNSBlock

Adds the following to the VLAN configuration:

```
vlan 915
  name 915_ISP1_PROVIDER
  apply access-list ip DNSBlock out
  apply access-list ipv6 DNSBlock out
  apply access-list mac DNSBlock out
```

[\(return to TOC\)](#)

# Create Role

Library > **Roles & Policies** > Roles

Create Role

Name \*

VTECH

Description

652\_GUEST\_VOIP

VLAN ID

Captive Portal Profile

Select

GPID \*

400

☐ Dynamic Application Prioritization

Device-Specific Parameters

☒ Switch

☐ Gateway

Switch Parameters

Authentication Mode

☒ Device

☐ Client

☐ Multidomain

VLAN

☐ Access

☒ Trunk

Native VLAN

1050

Allowed VLANs

652,1050

PoE

Critical

PoE Allocation

☐ Class

☒ Usage

MTU

☐ Admin Edge Port

Trust Mode

Dscp

Client Inactivity Timeout

Seconds

Reauthentication Period

Seconds

Cache Reauth Period

Seconds

Session Timeout

Seconds

☐ User Based Tunnel

Gateway Zone

Gateway Role

Select

Create

Assign to proper Device Function(s) and Scope(s)

Profiles

Roles & Policies

Named Objects

Services

Library

>

Roles

Q Search

Create Role

26 items

| Name                                                                       | GPID | Assigned Device Function                                                      | Assigned Scope | Referenced By |
|----------------------------------------------------------------------------|------|-------------------------------------------------------------------------------|----------------|---------------|
| <div><div><div></div></div><div>VTECH</div><div>652_GUEST_VOIP</div></div> | 400  | <div>Access Switch</div> <div>Aggregation Switch</div> <div>Core Switch</div> | Global         | 1 policy      |

Limitations on ‘Trunk’ Allowed VLAN quantity in Roles:

| Switch       | Named VLANs | Total VLANs (Including Named) |
|--------------|-------------|-------------------------------|
| CX-6300      | 50          | 1024                          |
| CX-6200      | 50          | 256                           |
| CX-6000/6100 | 50          | 50                            |

Currently, there is not a GUI option to set in the role:  
private-vlan port-secondary (must use API)

To modify the existing Role to change the VLAN to Private Type (SECONDARY, or PROMISCUOUS), use the API :

*PATCH {{cluster\_url}}/network-config/v1alpha1/roles/{{role\_name}}*

The JSON body example for PRIMARY VLAN is below (modify to your settings)

```
PATCH {{cluster_url}}/network-config/v1alpha1/roles/6300-user

{
  "vlan-parameters": {
    "private-vlan-port-type": "SECONDARY",
    "access-vlan": 13
  }
}
```

Adds the following to the Switch configuration:

```
port-access role 6300-user
  description Sample user with wired allow all policy
  associate policy sys_policy_pap_6300-user
  auth-mode device-mode
  client-inactivity timeout none
  poe-allocate-by usage
  private-vlan port-type secondary
  vlan access 13
```

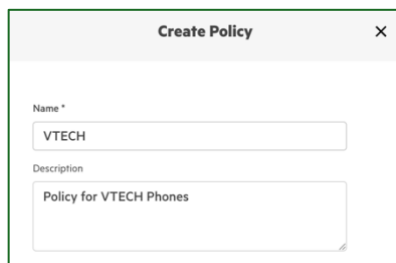
[\*\(return to TOC\)\*](#)

## Create Security Policies

**\* You must create and assign at least one Security Policy with a Role**

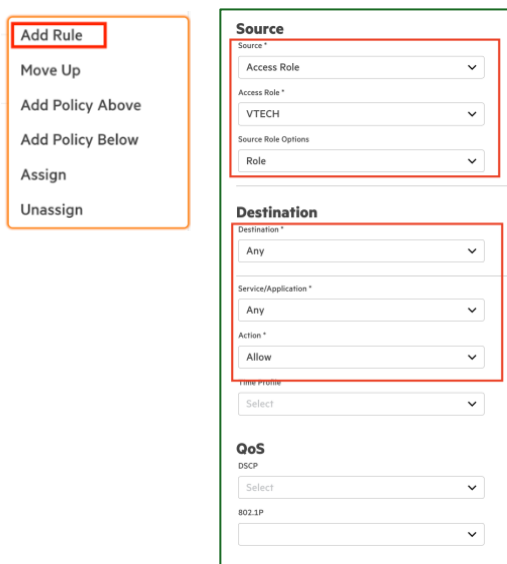
Library > **Roles & Policies** > Security Policies > Role-based Policies

### Create Policy



The 'Create Policy' dialog box has a title bar with a close button. It contains two text input fields: 'Name \*' with the value 'VTECH' and 'Description' with the value 'Policy for VTECH Phones'.

### Add rules to Policy



The 'Add Rule' dialog box has a title bar with a close button. It contains a list of actions: 'Add Rule', 'Move Up', 'Add Policy Above', 'Add Policy Below', 'Assign', and 'Unassign'. The 'Add Rule' button is highlighted with a red border. To the right of the dialog box is a form for adding a rule. The form has two main sections: 'Source' and 'Destination'. The 'Source' section has three dropdown menus: 'Access Role' (set to 'VTECH'), 'Access Role \*' (set to 'VTECH'), and 'Source Role Options' (set to 'Role'). The 'Destination' section has three dropdown menus: 'Destination \*' (set to 'Any'), 'Service/Application \*' (set to 'Any'), and 'Action \*' (set to 'Allow'). Below these are two more dropdown menus: 'Time Profile' (set to 'Select') and 'QoS' (set to 'Select').

*\*When modifying the Source (Role), the policy is associated with the role*

The example creates a rule to allow all traffic for the Role VTECH

| VTECH       |             | 1                   | Access Switch, Aggregation Switch, Core Switch |             |      | Global |
|-------------|-------------|---------------------|------------------------------------------------|-------------|------|--------|
| Source      | Destination | Service/Application | Action                                         | Description | DSCP | 802.1P |
| Role: VTECH | Any         | Any                 | Allow                                          | -           | -    | -      |

Add the following to the switch configuration:

```
class ip sys_pap_VTECH_ip_allow_1
  10 match any any any count
class ipv6 sys_pap_VTECH_ipv6_allow_1
  10 match any any any count

port-access policy sys_policy_pap_VTECH
  10 class ip sys_pap_VTECH_ip_allow_1
  20 class ipv6 sys_pap_VTECH_ipv6_allow_1

port-access role VTECH
  description 652_GUEST_VOIP
  associate policy sys_policy_pap_VTECH
  auth-mode device-mode
  poe-allocate-by usage
  poe-priority critical
  trust-mode dscp
  vlan trunk native 652
  vlan trunk allowed 652,1050
```

Example of Role (6300-user) with block to RFC-1918 Addresses:

To see the policies associated with a role:

The screenshot shows the 'Roles & Policies' section of the network management interface. The '6300-user' role is selected, and the 'References' tab is active, showing a list of policies: 'block\_rfc1918' with 3 rules and 'wired\_allowall' with 1 rule. A blue arrow points from the role name to the policy list.

| Name                                                                                        | Rules                   | Assigned Device Function                       | Assigned Scope                                                                            |                    |             |               |
|---------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------|-------------|---------------|
| <div><div>▼</div><div><b>block_rfc1918</b><br/>Blocks RFC 1918 addresses</div></div>        | 3                       | Access Switch, Core Switch                     | Global                                                                                    |                    |             |               |
| <b>Source</b>                                                                               | <b>Destination</b>      | <b>Service/Application</b>                     | <b>Action</b>                                                                             | <b>Description</b> | <b>DSCP</b> | <b>802.1P</b> |
| Role: 6300-user<br>Role                                                                     | Network: 192.168.0....  | Any                                            |  Deny    | -                  | -           | -             |
| Role: 6300-user<br>Role                                                                     | Network: 172.16.0.0/... | Any                                            |  Deny    | -                  | -           | -             |
| Role: 6300-user<br>Role                                                                     | Network: 10.0.0.0/8     | Any                                            |  Deny    | -                  | -           | -             |
| <div><div>▼</div><div><b>wired_allowall</b><br/>Policy to allow all traffic Use</div></div> | 2                       | Access Switch, Aggregation Switch, Core Switch | Global                                                                                    |                    |             |               |
| <b>Source</b>                                                                               | <b>Destination</b>      | <b>Service/Application</b>                     | <b>Action</b>                                                                             | <b>Description</b> | <b>DSCP</b> | <b>802.1P</b> |
| Role: 6300-user<br>Role                                                                     | Any                     | Any                                            |  Allow | -                  | -           | -             |

Add the following to the switch configuration:

```

gbp role 6300-user 700

class ip sys_pap_6300-user_ip_deny_1
  10 match any any 192.168.0.0/255.240.0.0 count
  20 match any any 172.16.0.0/255.255.0.0 count
  30 match any any 10.0.0.0/255.0.0.0 count

class ipv6 sys_pap_6300-user_ipv6_allow_4
  10 match any any any count

port-access policy sys_policy_pap_6300-user
  10 class ip sys_pap_6300-user_ip_deny_1 action drop
  20 class ip sys_pap_6300-user_ip_allow_4
  30 class ipv6 sys_pap_6300-user_ipv6_allow_4

port-access role 6300-user
  description Sample user with wired allow all policy
  associate policy sys_policy_pap_6300-user
  auth-mode device-mode
  client-inactivity timeout none
  poe-allocate-by usage
  vlan access 300

```

[\(return to TOC\)](#)

# Interface Settings

## Routed IP Interface

Select **Device** > Interfaces > Switch Interface Configuration

Choose Interface

Update with any additional settings that may be required (VRF, QoS, etc.)

| Port                                              | Admin Status | Speed/Duplex | Port Profile | LAG | Port Mode | Routing  |
|---------------------------------------------------|--------------|--------------|--------------|-----|-----------|----------|
| <input checked="" type="checkbox"/> <b>1/1/10</b> | Enabled      | Auto         | -            | -   | -         | Enabled  |
| <input type="checkbox"/> <b>1/1/11</b>            | Enabled      | Auto         | -            | 2   | -         | Disabled |
| <input type="checkbox"/> <b>1/1/12</b>            | Enabled      | Auto         | -            | 2   | -         | Disabled |

Add the following to the switch configuration:

```
interface 1/1/10
  no shutdown
  routing
  ip address 10.4.2.1/24
  exit
```

## Create VLAN

Profiles > Library > VLANs & Networks > VLAN > Create Profile

The image shows three screenshots of the VLAN configuration interface:

- Edit Profile:** Shows the 'Properties' tab with fields for VLAN ID (100), Name (100\_LAN\_Management), and Description. It includes a 'Switch Specific Parameters' section with 'Enable VLAN' checked. The 'Switch Parameters' section includes 'Enable L3' (checked), 'VRF Profile' (None), 'Policy' (None), and various checkboxes for Access List, DHCP v4/v6 Snooping, Enable ARP Inspection, Voice, and Enable IP Client Tracker.
- Assign Profile:** Shows the 'Device Function' section with 'Access Switch', 'Aggregation Switch', 'Core Switch', and 'Mobility Gateway' all checked. The 'Scopes' section shows 'Name' and 'Scope Level' both set to 'Global'.
- Summary View:** Shows the 'Enable L3' checkbox checked, 'IPv4 Alias' unchecked, 'IPv4 Address Assignment' set to 'Static', 'IPv6 Address Assignment' set to 'Static', 'Relay to External' unchecked, 'Admin State' checked, 'MTU' field, 'IGMP' set to 'Disabled', 'VRRP Router' set to 'None', and 'Switch Specific Parameters' and 'Gateway Specific Parameters' sections.

\* **Note:** Enable L3 if SVI is required (interface vlan)

Adds the following to the switch configuration:

```
vlan 100
  name 100_LAN_Managment
```

Repeat for each VLAN required

[\(return to TOC\)](#)

## Create Private VLAN (PVLAN)

*\* At this time, Private VLANs can only be configured using the APIs*

Create the Primary VLAN, then add Isolated/community as needed

The API call to Create (POST) the VLAN is:

**POST** `{{cluster_url}}/network-config/v1alpha1/layer2-vlan/{{vlanid}}`

The JSON body example for PRIMARY VLAN is below (modify to your settings)

```
POST {{cluster_url}}/network-config/v1alpha1/layer2-vlan/11

{
  "private-vlan-type": "PRIMARY",
  "name": "11_PVLAN_11",
  "vlan": 11
  "voice-enable": false,
  "enable": true,
  "client-ip-tracker-enable": true,
  "dhcpv4-snooping": {
    "enable": false
  },
  "dhcpv6-snooping": {
    "enable": false
  }
}
```

The JSON body example for ISOLATED VLAN is below (modify to your settings)

```
POST {{cluster_url}}/network-config/v1alpha1/layer2-vlan/12

{
  "private-vlan-type": "ISOLATED",
  "name": "12_PVLAN_12",
  "vlan": 12
  "voice-enable": false,
  "enable": true,
  "client-ip-tracker-enable": true,
  "dhcpv4-snooping": {
    "enable": false
  },
  "dhcpv6-snooping": {
    "enable": false
  }
}
```



The JSON body example for COMMUNITY VLAN is below (modify to your settings)

```
POST {{cluster_url}}/network-config/v1alpha1/layer2-vlan/31

{
  "private-vlan-type": "COMMUNITY",
  "name": "13_PVLAN_13",
  "vlan": 13
  "voice-enable": false,
  "enable": true,
  "client-ip-tracker-enable": true,
  "dhcpv4-snooping": {
    "enable": false
  },
  "dhcpv6-snooping": {
    "enable": false
  }
}
```

Assign each VLAN to correct Device Function and Scope

| VLAN ID | Name       | Assigned Device Function | Assigned Scope |
|---------|------------|--------------------------|----------------|
| 11      | 11_PVLAN11 | Access Switch            | Global         |
| 12      | 12_PVLAN12 | Access Switch            | Global         |
| 13      | 13_PVLAN13 | Access Switch            | Global         |

Adds the following to the switch configuration:

```
vlan 11
  name 11_PVLAN11
  client track ip
  private-vlan primary

vlan 12
  name 12_PVLAN12
  client track ip
  private-vlan community primary-vlan 11

vlan 13
  name 13_PVLAN13
  client track ip
  private-vlan isolated primary-vlan 11
```

[\(return to TOC\)](#)

## Create SVI – VLAN Interface

See section [Create Aliases](#) above if using for IP assignment

Select Profiles > **Device** > VLANs & Networks > VLAN

Edit the VLAN that requires an IP Address (MUST be configured L3)

Edit Profile

PropertiesReferences

This profile is overriding GLOBAL

VLAN ID \*

100

Name

100\_LAN\_Managment

This configuration is only applicable to Switches

Description

Switch Specific Parameters

Enable L3

IPv4 Alias

IPv4 Address Assignment \*

Static

DHCP

CX device doesn't support configuring DHCP-Client on more than one VLAN

IPv4 Address/Prefix

10.4.3.130/25

IPv6 Address Assignment \*

Static

DHCP

CX device doesn't support configuring DHCP-Client on more than one VLAN

IPv6 Address/Prefix

xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx/xxx

Switch Specific Parameters

Switch Parameters

Enable VLAN

VRF Profile

Policy

None

Access List

DHCP v4 Snooping

DHCP v6 Snooping

Enable ARP Inspection

Voice Enable Alias

Voice

Enable IP Client Tracker

IGMP

Enable IGMP Snooping

Advanced

MLD

Enable MLD Snooping

Advanced

Select “Switch Specific Parameters” to associate L3 with a different VRF, enable policy, and other settings

Shows profile assigned to local device – overrides global

| VLAN ID | Name              | Inherits From | Assigned Device Function | Assigned Scope | Overrides |
|---------|-------------------|---------------|--------------------------|----------------|-----------|
| 100     | 100_LAN_Managment | Self          | Access Switch            | TW11KNP03V     | -         |

Adds the following to the switch configuration:

```
interface vlan 100
  ip address 10.4.3.130/25
```

Repeat for each SVI required

[\(return to TOC\)](#)

Version 2025.1105

DRAFT

40

## Create LAG

Select Profiles > **Device** > Interfaces > Switch Interface Configuration  
Select ports then Create LAG

Profiles Roles & Policies Named Objects

DCARW-MDF-SW01 > Interfaces > **Switch Interface Configuration** Core Switch

Ports LAGs

Search

**Create LAG** Apply Port Profile

28 items

DCARW-MDF-SW01

CX-6300M

Legend

- Connected
- Error
- Disabled
- Not Connected
- Uplink
- Delivering PoE

Select Information to Display

Neighbor Type

Switch

| Port                                       | Admin Status | Speed/Duplex | Port Profile | LAG | Port Mode | Routing  |
|--------------------------------------------|--------------|--------------|--------------|-----|-----------|----------|
| <input checked="" type="checkbox"/> 1/1/11 | Enabled      | Auto         | -            | -   | Access    | Disabled |
| <input checked="" type="checkbox"/> 1/1/12 | Enabled      | Auto         | -            | -   | Access    | Disabled |
| <input type="checkbox"/> 1/1/13            | Enabled      | Auto         | -            | -   | Access    | Disabled |

Create LAG

**LAG**

Configure Link Aggregation (LAG) for ports.

Set IP Tracker to 'Auto' when creating a LAG for use as an Inter-Switch Link (ISL) between VSX peers.

**General**

Ports

1/1/11

1/1/12

1-480

2

Description

Lag to IDF Switch

☒ Enable

Type \*

LAG

MCLAG

Mode \*

Static

LACP

LACP Mode \*

Active

Rate \*

Slow (Default)

☐ LACP Fallback Static

**Network Parameters**

☐ Routing

VLAN Mode \*

Access

Trunk

Native VLAN \*

100

☐ Tag

Allowed VLANs \*

all

☐ DHCPv4 Snooping Trust

Max Binding

☐ DHCPv6 Snooping Trust

Max Binding

☐ ARP Inspection Trust

☐ Block Port Until Device Profile Active

**Policies**

Policy

Select

☐ Access List

**QoS**

QoS Trust

Select

☐ Shape Outbound Traffic

**Loop Prevention**

☐ Loop Protection

STP Port-Priority

0 (Highest) 15 (Lowest)

4

STP Port-Cost

STP Options

☐ BPDU Filter

☐ BPDU Guard

☐ Admin Edge

☐ Root Guard

☐ Loop Guard

☐ TCN Guard

☐ RPVST Filter

☐ RPVST Guard

**Rate limit**

☐ Broadcast

☐ Multicast

☐ Unknown Unicast

☐ ICMP

**Security**

☐ IPv4 Source Lockdown

☐ IPv6 Source Lockdown

Not all options are displayed in the LAG configuration above  
Adjust to your specific requirements

Adds the following to the switch configuration:

```
interface lag 2
  description Lag to IDF Switch
  no shutdown
  no routing
  vlan trunk native 100
  vlan trunk allowed all
  lacp mode active
  no aaa authentication port-access allow-lldp-auth
  no aaa authentication port-access allow-cdp-auth
  exit

interface 1/1/11
  no shutdown
  lag 2
  exit

interface 1/1/12
  no shutdown
  lag 2
  exit
```

[\(return to TOC\)](#)

## Device Profile

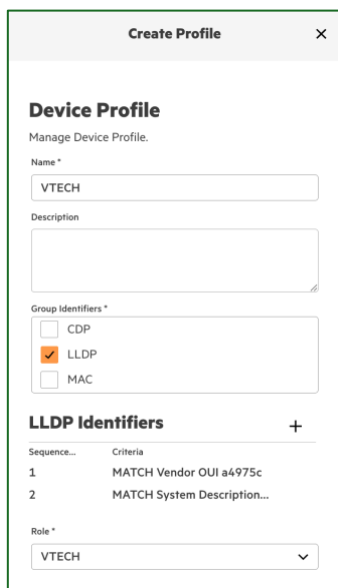
Profiles > Library > Interface > Device Profile

To create a Device Profile, a [Role](#) and [Security Policies](#) must be created

### Create Interface Device Profile

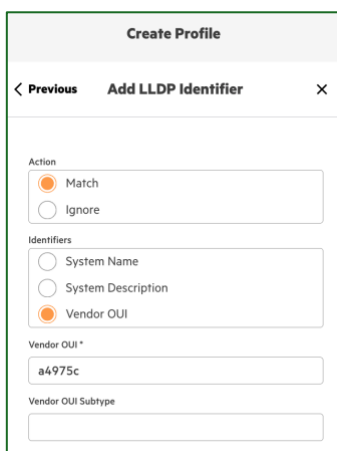
To use the LLDP Group, you must create a Role to be assigned

You can create the role within the workflow, or using the instructions above

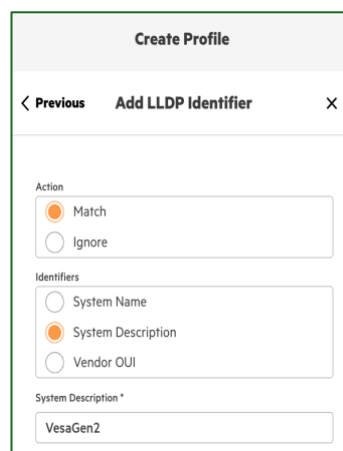


The 'Create Profile' form for a Device Profile. It includes fields for Name (VTECH), Description, and Group Identifiers (CDP, LLDP, MAC). The LLDP checkbox is checked. Below, the 'LLDP Identifiers' section shows a sequence of two criteria: 'MATCH Vendor OUI a4975c' and 'MATCH System Description...'. A Role dropdown is set to VTECH.

- Select the Group Identifiers (CDP, LLDP, MAC)
- Specify LLDP Identifiers and add in sequence
- Select/Create Role
- Assign Profile to Device Function(s) & Scope(s)



The 'Add LLDP Identifier' form. It shows Action (Match selected), Identifiers (Vendor OUI selected), Vendor OUI (a4975c), and Vendor OUI Subtype.



The 'Add LLDP Identifier' form. It shows Action (Match selected), Identifiers (System Description selected), and System Description (VesaGen2).

Assign to proper Device Function(s) and Scope(s)

| Profiles                              | Roles & Policies | Named Objects                                  | Services       |
|---------------------------------------|------------------|------------------------------------------------|----------------|
| Library > Interfaces > Device Profile |                  |                                                |                |
| Q Search                              |                  |                                                |                |
| 1 item                                |                  |                                                |                |
| Name                                  | Roles            | Assigned Device Function                       | Assigned Scope |
| > VTECH                               | VTECH            | Core Switch, Aggregation Switch, Access Switch | Global         |

Adds the following to the switch configuration:

```
port-access lldp-group lldp_grp_VTECH
  seq 1 match vendor-oui a4975c
  seq 2 match sys-desc VesaGen2

port-access device-profile VTECH
  enable
  associate role VTECH
  associate lldp-group lldp_grp_VTECH
```

## Example for Aruba Devices

Using the Subtype and Value will help limit the LLDP MATCH to Aruba APs, CX-Switches, and Gateways connected to the switch.

\* At this time “*vendor-oui-subtype value*” can only be configured using the APIs

To specify the LLDP Match criteria for Aruba Devices, use the following:

Aruba GW: vendor-oui: 000B86; subtype: type 1, value 0002  
Aruba AP: vendor-oui: 000B86; subtype: type 1, value 0001  
Aruba CX: vendor-oui: 883A30; subtype: type 2, value 0001  
Aruba AOSS: vendor-oui: 0016B9; subtype: type 2, value 0001

To determine what Vendor OUI, Subtype, and Values, issue the command from the CX Switch CLI:

```
CX-Switch# show lldp oui-tlv-details
LOCAL-PORT  CHASSIS-ID          PORT-ID    OUI      OUI-SUBTYPE  OUI-DATA
-----
1/1/1        20:4c:03:d3:b1:4e    20:4c:0... 000b86    1            0001
1/1/1        20:4c:03:d3:b1:4e    20:4c:0... 000b86    4            0000
...
1/1/13       88:3a:30:96:f8:40    1/1/1       883a30    2            0001
...
1/1/14       20:4c:03:5f:7c:e2    GE0/0/1     000b86    1            0002
...
1/1/23       f8:60:f0:c8:65:8d    9           0016b9    2            0001
```

The API to Create the Device Profile to include the “*vendor-oui-subtype: value*” is:  
**POST** `{{cluster_url}}/network-config/v1alpha1/device-profile/{{profile_name}}`

The JSON body examples are below (modify to your settings)

Aruba Access Point (*profile\_name = arubaAP*)

```
POST {{cluster_url}}/network-config/v1alpha1/device-profile/{{profile_name}}

{
  "name": "arubaAP",
  "role": "arubaAP",
  "lldp-group-entries": [
    {
      "sequence-number": 1,
      "action": "MATCH",
      "vendor-oui": "000B86",
      "vendor-oui-subtype": [
        {
          "type": 1,
          "value": "0001"
        }
      ]
    }
  ]
}
```

Aruba Gateway (*profile\_name = arubaGW*)

```
POST {{cluster_url}}/network-config/v1alpha1/device-profile/{{profile_name}}

{
  "name": "arubaGW",
  "role": "arubaGW",
  "lldp-group-entries": [
    {
      "sequence-number": 1,
      "action": "MATCH",
      "vendor-oui": "000B86",
      "vendor-oui-subtype": [
        {
          "type": 1,
          "value": "0002"
        }
      ]
    }
  ]
}
```

Aruba Switch (*profile\_name = arubaSW*)

**POST** `{{cluster_url}}/network-config/v1alpha1/device-profile/{{profile_name}}`

```
{
  "name": "arubaSW",
  "role": "arubaSW",
  "lldp-group-entries": [
    {
      "sequence-number": 1,
      "action": "MATCH",
      "vendor-oui": "883A30",
      "vendor-oui-subtype": [
        {
          "type": 2,
          "value": "0001"
        }
      ]
    },
    {
      "sequence-number": 2,
      "action": "MATCH",
      "vendor-oui": "0016B9",
      "vendor-oui-subtype": [
        {
          "type": 2,
          "value": "0001"
        }
      ]
    }
  ]
}
```

name = Device Profile

role = Role to be assigned if LLDP options Match

The above examples add the following to the switch configuration:

```
port-access lldp-group lldp_grp_arubaAP
  seq 1 match vendor-oui 000B86 type 1 value 0001

port-access lldp-group lldp_grp_arubGW
  seq 1 match vendor-oui 000B86 type 1 value 0002

port-access lldp-group lldp_grp_arubaSW
  seq 1 match vendor-oui 000B86 type 2 value 0001
  seq 2 match vendor-oui 0016B9 type 2 value 0001
```

[\(return to TOC\)](#)



## Basic Interface Setting

Interface settings are accomplished either per-interface or by using a Port Profile. If you choose per-interface, each interface must be configured for the settings to be deployed.

Example **Interface** Config: Underlay Access Point – NO AUTHENTICATION

Select Profiles > **Device** > Interfaces > Switch Interface Configuration

Choose the interface and apply individual settings that are required for the port

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Port *<br/>1/1/4</p> <p>Description<br/>AP - Underlay</p> <p><input type="checkbox"/> Use Port Profile</p> <p><input checked="" type="checkbox"/> Enable Port</p> <p>Speed Duplex<br/>Auto</p> <p>MTU<br/>4000</p> <p><b>UDLD</b><br/><input type="checkbox"/> Enable UDLD</p> <p><b>LLDP and CDP</b><br/>LLDP Mode<br/>Transmit and Receive</p> <p><input checked="" type="checkbox"/> Enable CDP</p> <p><b>Network Parameters</b><br/><input type="checkbox"/> Routing</p> <p>VLAN Mode *<br/><input type="radio"/> Access<br/><input checked="" type="radio"/> Trunk</p> <p>Native VLAN *<br/>101</p> <p><input type="checkbox"/> Tag</p> <p>Allowed VLANs *<br/>101,300,1000,1001</p> <p><input type="checkbox"/> DHCPv4 Snooping Trust</p> <p>Max Binding<br/><input type="checkbox"/> DHCPv6 Snooping Trust</p> <p>Max Binding<br/><input type="checkbox"/> ARP Inspection Trust</p> <p><input type="checkbox"/> Block Port Until Device Profile Active</p> | <p><b>PoE</b><br/><input checked="" type="checkbox"/> Enable PoE</p> <p>Priority *<br/>High</p> <p>PoE Allocation By *<br/>Usage</p> <p><input type="checkbox"/> Pre-Standard Detect</p> <p><b>Policies</b><br/>Policy<br/>None</p> <p><input type="checkbox"/> Access List</p> <p><b>QoS</b><br/>QoS Trust<br/>DSCP</p> <p>Specify DSCP Value<br/>0-63</p> <p><input type="checkbox"/> Shape Outbound Traffic</p> <p><b>Loop Prevention</b><br/><input checked="" type="checkbox"/> Loop Protection</p> <p>VLANs</p> <p>Action<br/>Select</p> <p>STP Port-Priority<br/>0 (Highest) 15 (Lowest)<br/>8</p> <p>STP Port-Cost</p> <p>STP Options<br/><input type="checkbox"/> BPDU Filter<br/><input type="checkbox"/> BPDU Guard<br/><input checked="" type="checkbox"/> Admin Edge<br/><input type="checkbox"/> Root Guard<br/><input type="checkbox"/> Loop Guard<br/><input type="checkbox"/> TCN Guard<br/><input type="checkbox"/> RPVST Filter<br/><input type="checkbox"/> RPVST Guard</p> | <p><b>Rate limit</b><br/><input type="checkbox"/> Broadcast<br/><input type="checkbox"/> Multicast<br/><input type="checkbox"/> Unknown Unicast<br/><input type="checkbox"/> ICMP</p> <p><b>Security</b><br/><input type="checkbox"/> IPv4 Source Lockdown<br/><input type="checkbox"/> IPv6 Source Lockdown</p> <p><b>Authentication</b><br/>Authentication Protocol<br/>None</p> <p>Client Limit</p> <p><input type="checkbox"/> Allow Flood Traffic</p> <p>Authentication Mode<br/>Select</p> <p>Client Auto Logout<br/><input type="checkbox"/> Final Authentication Failure</p> <p>Onboarding Precedence<br/>Select</p> <p><b>LLDP Support</b><br/><input type="checkbox"/> Allow LLDP Authentication<br/><input type="checkbox"/> Allow LLDP Bypass</p> <p><b>CDP Support</b><br/><input type="checkbox"/> Allow CDP Authentication<br/><input type="checkbox"/> Allow CDP Bypass<br/><input type="checkbox"/> Allow CDP Proxy-logout</p> <p><b>Security Violation</b><br/>Action<br/>Select</p> <p><b>Fault Monitoring</b><br/><input type="checkbox"/> Enable Fault Monitoring</p> <p><b>Port Security</b><br/><input type="checkbox"/> Enable Port Security</p> <p><b>Telemetry</b><br/>IP Client Tracker<br/>Auto</p> <p>IP Tracking Client Limit</p> <p><input type="checkbox"/> Enable Application Recognition</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Adds the following to the switch configuration:

```
interface 1/1/4
  description AP - Underlay
  no shutdown
  mtu 4000
  no routing
  vlan trunk native 101
  vlan trunk allowed 101,300,1000-1001
  qos trust dscp
  spanning-tree port-type admin-edge
  power-over-ethernet priority high
  loop-protect
  exit
```

*\*Verify the MTU required (example displayed)*

To apply Rate Limiting, update the **Rate limit** section on the individual interface (or port profile) with desired settings.

**Rate limit**

☒ Broadcast

Unit: %

10

☒ Multicast

Unit: %

12

☐ Unknown Unicast

☐ ICMP

Adds the following to the switch configuration:

```
interface 1/1/4
  ...
  rate-limit broadcast 10 percent
  rate-limit multicast 12 percent
  ...
```

[\(return to TOC\)](#)

## Interface Authentication Configuration

To enable Authentication, accomplish the following tasks prior to interface config:

[Enable Global Authentication](#)

[Create AAA Server\(s\)](#)

[Create AAA Server Group](#)

Example **Interface** Config (Authentication Option):

Profiles > **Device** > Interface > Switch Interface Configuration

Select Interface

Navigate to the Authentication Section and update with desired settings:

The image displays three panels from a network configuration interface:

- Authentication:** Includes fields for Authentication Protocol (set to Concurrent), Client Limit (set to 2), Allow Flood Traffic (unchecked), Authentication Mode (set to Client mode), Client Auto Logoff (unchecked), Final Authentication Failure (unchecked), and Onboarding Precedence (set to Device Profile then AAA).
- Authorization Parameters:** Includes dropdown menus for Select Critical Auth Role, Select Pre Auth Role, Select Reject Auth Role, Select VOIP Auth Role, Select Authentication Role, and Select Fallback Role, each with a description of the role's application.
- 802.1X Parameters:** Includes input fields for Max Authentication Failure, Reauthentication Interval, Cached Reauthentication Period, Discovery Period, EAPOL Timeout, Max EAPOL Requests, and Quiet Period, each with a unit of seconds.

Adds the following to the switch configuration:

```
interface 1/1/2
  no shutdown
  no routing
  vlan access 1000
  spanning-tree port-type admin-edge
  port-access onboarding-method precedence device-profile aaa
  port-access onboarding-method concurrent enable
  no aaa authentication port-access allow-lldp-auth
  no aaa authentication port-access allow-cdp-auth
  aaa authentication port-access client-limit 2
  aaa authentication port-access dot1x authenticator
  enable
  aaa authentication port-access mac-auth
  enable
  loop-protect
  exit
```

To apply Rate Limiting, update the **Rate limit** section on the individual interface (or port profile) with desired settings.

**Rate limit**

☒ Broadcast

Unit  
%

10

☒ Multicast

Unit  
%

12

☐ Unknown Unicast

☐ ICMP

Adds the following to the switch configuration:

```
interface 1/1/12
...
rate-limit broadcast 10 percent
rate-limit multicast 12 percent
...
```

[\(return to TOC\)](#)

## Port Profile Configuration

**Port Profiles** allow common settings to be grouped together and applied to multiple interfaces with a simple configuration step

Profiles > Library > Interfaces > Port Profiles

Example Port Profile Config: Underlay Access Point – **NO AUTHENTICATION**

Create Port Profile and apply settings that are required for the ports  
(will be common on all ports that are configured with the Port Profile)

The screenshot displays the 'Port Profile Configuration' interface with the following settings:

- General:**
  - ☐ Create as a local profile
  - Name: ap\_underlay
  - Description: AP - Underlay Trunk
  - Device: Switch
  - ☒ Admin State
  - Speed/Duplex: Auto
  - Jumbo Framed MTU: 9198
- PoE:**
  - ☒ Enable PoE
  - Priority: Critical
  - PoE Allocation By: Usage
  - ☐ Pre-Standard Detect
- UDLD:**
  - ☐ Enable UDLD
- LLDLP and CDP:**
  - LLDP Mode: Transmit and Receive
  - ☒ Enable CDP
- Network Parameters:**
  - ☐ Use Switchport Alias
  - VLAN Mode: Trunk
  - Native VLAN: 101
  - ☐ Tag
  - Allowed VLANs: 101,300,1000,1001
  - ☐ DHCPv4 Snooping Trust
  - Max Binding:
  - ☐ DHCPv6 Snooping Trust
  - Max Binding:
  - ☐ ARP Inspection Trust
- Loop Prevention:**
  - ☒ Loop Protection
  - VLANs:
  - Action: Transmit disable
  - STP Priority: 8 (slider from 0 to 15)
  - STP Cost:
  - STP Options:
    - ☒ Admin Edge
    - ☐ BPDU Guard
    - ☐ BPDU Filter
    - ☐ Root Guard
    - ☐ Loop Guard
    - ☐ TCN Guard
    - ☐ RPPVST Filter
    - ☐ RPPVST Guard
- Security:**
  - ☐ Enable Port Authentication
  - Policy: None
  - ☐ Access List
- Fault Monitoring:**
  - ☐ Enable Fault Monitoring
- QoS:**
  - QoS Trust: DSCP
- Advanced:**
  - Rate Limit:**
    - ☐ Broadcast
    - ☐ Multicast
    - ☐ Unknown Unicast
    - ☐ ICMP
  - Telemetry:**
    - IP Client Tracker: Auto
    - IP Tracking Client Limit:

Assign to Device Function & Scopes:

The 'Assign Profile' dialog box shows the following configuration:

- Device Function:**
  - ☒ Access Switch
  - ☒ Aggregation Switch
  - ☐ Bridge
  - ☒ Core Switch
  - ☐ Campus Access Point
  - ☐ Mobility Gateway
- Scopes:**
  - Name: Global
  - Scope Level: Global

Assign the Port Profile to the Interfaces:  
Select Profiles > **Device** > Interfaces > Switch Interface Configuration  
Choose the interface(s) and apply Port Profile settings:

Single Interface:

The screenshot shows the 'Configure Interface' dialog box. The 'Port' field is set to '1/1/5'. The 'Description' field is empty. The 'Use Port Profile' checkbox is checked. The 'Port Profile' dropdown is set to 'ap\_underlay'.

Multiple Interfaces:

The screenshot shows the 'Apply Port Profile' dialog box. The 'Ports' list includes '1/1/5', '1/1/6', '1/1/7', '1/1/8', and '1/1/9'. The 'Use Port Profile' checkbox is checked. The 'Port Profile' dropdown is set to 'ap\_underlay'. A warning message states: 'Port profile cannot be removed if the selected ports have dissimilar Port/Port profile configuration.'

| Port                                      | Admin Status | Speed/Duplex | Port Profile | LAG | Port Mode |
|-------------------------------------------|--------------|--------------|--------------|-----|-----------|
| <input checked="" type="checkbox"/> 1/1/5 | Enabled      | Auto         | -            | -   | Access    |
| <input checked="" type="checkbox"/> 1/1/6 | Enabled      | Auto         | -            | -   | Access    |
| <input checked="" type="checkbox"/> 1/1/7 | Enabled      | Auto         | -            | -   | Access    |
| <input checked="" type="checkbox"/> 1/1/8 | Enabled      | Auto         | -            | -   | Access    |
| <input checked="" type="checkbox"/> 1/1/9 | Enabled      | Auto         | -            | -   | Access    |

Adds the following to the switch configuration:

```
interface 1/1/5
  description AP - Underlay Trunk
  no shutdown
  mtu 9198
  no routing
  vlan trunk native 101
  vlan trunk allowed 101,300,1000-1001
  qos trust dscp
  spanning-tree port-type admin-edge
  power-over-ethernet priority critical
  loop-protect
  exit

interface 1/1/6
  ...
```

## Example Port Profile Config: Client Device – **WITH AUTHENTICATION**

To enable Authentication, accomplish the following tasks prior to interface config:

[Enable Global Authentication](#)

[Create AAA Server\(s\)](#)

[Create AAA Server Group](#)

[Create Authentication Profile](#)

Create Port Profile and apply settings that are required for the ports  
(will be common on all ports that are configured with the Port Profile)

The screenshot displays the 'Port Profile' configuration interface with several tabs and settings:

- General:** Includes fields for Name (port-auth-general), Description (Port Profile - General Authentication), Device (Switch), and Admin State (checked). It also has Speed/Duplex (Auto) and Jumbo Framed MTU (1500).
- Network Parameters:** Includes VLAN Mode (Access selected), Access VLAN (1000), and checkboxes for DHCPv4, DHCPv6, and ARP Inspection Trust.
- Loop Prevention:** Includes Loop Protection (checked), Action (Transmit disable), STP Priority (8), STP Cost, and STP Options (Admin Edge checked).
- PoE:** Includes Enable PoE (checked), Priority (Low), PoE Allocation By (Usage), and Pre-Standard Detect.
- UDLD:** Includes Enable UDLD.
- LLDP and CDP:** Includes LLDP Mode (Transmit and Receive) and Enable CDP (checked).
- Security:** Includes Enable Port Authentication (checked), AAA Profile (wired-port-auth-no-flood), Policy (None), and Access List.
- Fault Monitoring:** Includes Enable Fault Monitoring.
- QoS:** Includes QoS Trust (DSCP).
- Advanced:** Includes Rate Limit (Broadcast, Multicast, Unknown Unicast, ICMP).
- Telemetry:** Includes IP Client Tracker (Auto) and IP Tracking Client Limit.

Configure the Port Profile with the Enable Port Authentication and Profile selected

Assign to Device Function & Scopes:

The 'Assign Profile' dialog box shows the following configuration:

- Device Function:** Access Switch, Aggregation Switch, Core Switch, and Campus Access Point are checked. Bridge and Mobility Gateway are unchecked.
- Scopes:** A table with two columns: Name and Scope Level. The row shows 'Global' for both.



## Assign Port Profile to Interface(s)

Profiles > **Device** > Interfaces > Switch Interface Configuration

Configure Interface

Port \*

1/1/3

Description

☒ Use Port Profile

Port Profile \*

port-auth-general

| Port                           | Admin Status | Speed/Duplex | Port Profile      | LAG | Port Mode | Routing |
|--------------------------------|--------------|--------------|-------------------|-----|-----------|---------|
| <input type="checkbox"/> 1/1/3 | -            | -            | port-auth-general | -   | -         | -       |

Adds the following to the switch configuration:

```
interface 1/1/3
  description Port Profile - General Authentication
  no shutdown
  vlan access 1000
  qos trust dscp
  spanning-tree port-type admin-edge
  port-access onboarding-method precedence device-profile aaa
  port-access onboarding-method concurrent enable
  aaa authentication port-access allow-cdp-bpdu
  aaa authentication port-access allow-lldp-bpdu
  aaa authentication port-access client-limit 2
  aaa authentication port-access dot1x authenticator
    eapol-timeout 5
    radius server-group RADIUS-1-Group
    max-eapol-requests 1
    enable
  aaa authentication port-access mac-auth
    enable
    radius server-group RADIUS-1-Group
  loop-protect
  exit
```

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# Troubleshooting

## Port Mirroring

\* At this time, Port Mirroring can only be configured using the APIs

## Tools

The following tests are available in New Central and can be run from/to internal and external destinations

Navigate to <Site> select a device and click the tools icon:



Make selections for tests to run and view in the results section

example:

The screenshot displays the 'Tools' interface for a device named 'DCARW-MDF-SW01'. The 'Tests' tab is active, showing a list of available tests: Ping (selected), Traceroute, PoE Bounce, Port Bounce, Cable Test, HTTP Test, Route Table, MAC Table, ARP Table, and AAA Test. The 'Destination Type' is set to 'IP/Hostname', and the 'Destination' is '192.168.100.1'. The 'Use IPv6' toggle is off. Under 'Show Additional Settings', 'Use Management Interface' is on, and 'VRF Name' is empty. 'Packet size' is set to '1000' and 'Count' is '10'. The 'Results' section shows a successful Ping test completed at 20:05:58. It includes a table of inputs (Device: DCARW-MDF-SW01, Destination: 192.168.100.1, Packet size: 1000 Bytes, Count: 10) and outputs (Packet Loss: 0%, Average Round Trip: 0.514 ms). A detailed log shows 10 successful ping attempts with round trip times ranging from 0.463 ms to 0.538 ms. The final statistics show 10 packets transmitted, 10 received, 0% packet loss, and a total time of 9193ms.

```
PING 192.168.100.1 (192.168.100.1) 1000(1028) bytes of data.
1008 bytes from 192.168.100.1: icmp_seq=1 ttl=253 time=0.538 ms
1008 bytes from 192.168.100.1: icmp_seq=2 ttl=253 time=0.494 ms
1008 bytes from 192.168.100.1: icmp_seq=3 ttl=253 time=0.506 ms
1008 bytes from 192.168.100.1: icmp_seq=4 ttl=253 time=0.490 ms
1008 bytes from 192.168.100.1: icmp_seq=5 ttl=253 time=0.497 ms
1008 bytes from 192.168.100.1: icmp_seq=6 ttl=253 time=0.694 ms
1008 bytes from 192.168.100.1: icmp_seq=7 ttl=253 time=0.484 ms
1008 bytes from 192.168.100.1: icmp_seq=8 ttl=253 time=0.510 ms
1008 bytes from 192.168.100.1: icmp_seq=9 ttl=253 time=0.466 ms
1008 bytes from 192.168.100.1: icmp_seq=10 ttl=253 time=0.463 ms

--- 192.168.100.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9193ms
rtt min/avg/max/mdev = 0.463/0.514/0.694/0.063 ms
```

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## Console Access

Navigate to <Site> select a device and click the tools icon:



Select Remote Console

Enter Username, Password and start session

### Troubleshoot

For DCARW-MDF-SW01

**Tests** Remote Console Commands History

#### Sessions

Username \*

Password \*

Cancel Start Session

### Troubleshoot

For DCARW-MDF-SW01

**Tests** Remote Console Commands History

#### Sessions

New Session

admin@DCARW-MDF-SW01  
SSH session started

#### Console

Search

Last login: 2025-10-27 23:05:07 from 192.168.10.252  
User "admin" has logged in 10 times in the past 30 days  
DCARW-MDF-SW01# show version  
-----  
ArubaOS-CX  
(c) Copyright 2017-2025 Hewlett Packard Enterprise Development LP  
-----  
Version : FL.10.13.1110  
Build Date : 2025-06-16 23:54:14 UTC  
Build ID : ArubaOS-CX:FL.10.13.1110:40649b64b204:202506162316  
Build SHA : 40649b64b2043235df776e64a158f0f58842c868  
Hot Patches :  
Active Image : secondary  
  
Service OS Version : FL.01.14.0002  
BIOS Version : FL.01.0002  
DCARW-MDF-SW01#

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## FAQs

How do I configure a stack using ZTP?

How do I configure LAG before device connects to network?

What are the CX Authentication Role types?

- **Critical Auth Role**  
Role applied when authentication server is not reachable
- **Pre Auth Role**  
Role applied for clients that are not authenticated
- **Reject Role**  
Role applied for clients failing authentication
- **VOIP Auth Role**  
Role applied for VOIP clients when authentication server is not reachable
- **Auth Role**  
Role applied for authenticated clients without roles assigned by authentication server
- **Fallback Role**  
Role applied for clients when no role is assigned via any of the onboarding method
- **UBT Fallback Role**  
Role applied for clients when UBT zone is not reachable

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